

Database Project Proposal Form

Project #		
Project Title:	Smart Hospital Management System with Integrated Network Infrastructure	
TA Name:	Eng. Mohamed Saeed	
Students' names:	<i>ID Number</i>	<i>Name</i>
1-	24030150	Eyad Ayman
2-	24030126	Youssef Sameh Ahmed
3-	24030055	Moatasem Mohamed Mostafa
4-	24030005	Ahmed Hesham Ramadan
5-	24030006	Marwan Bahaa Mohamed
6-	24030217	Youssef Wael Atef
Statement of Problem:	Traditional hospital management systems often rely on manual processes or fragmented digital solutions, leading to data redundancy, delays in data retrieval, and lack of real-time communication between departments. Additionally, the absence of an integrated network-based system makes it difficult to monitor patients remotely and coordinate efficiently between doctors, nurses, and administrative staff. Therefore, there is a need for a centralized smart system that integrates database management with network technologies to enhance data accuracy, accessibility, and communication.	
Brief Description of the Project:	The Smart Hospital Management System is a database-driven application designed to manage hospital operations efficiently. The system will handle patient records, doctor schedules, appointments, medical history, and billing information. It will also integrate networking concepts to enable real-time data sharing between different hospital departments and support remote monitoring. The system is intended for use by hospital staff such as doctors, nurses, and administrators, providing a centralized and secure platform for managing hospital data.	
Objectives:	- To design a centralized database for storing hospital data efficiently - To reduce data redundancy and improve data consistency - To enable fast and accurate data retrieval	

	• To facilitate real-time communication between hospital departments using network concepts • To improve patient management and service quality • To ensure data security and controlled access for different users
Technical Approach:	The system will be implemented using a relational database management system such as MySQL or SQL Server. An Entity-Relationship Diagram (ERD) will be designed to model the system entities such as Patients, Doctors, Appointments, and Departments. The database will consist of multiple related tables with proper normalization to ensure data integrity. SQL queries will be used for data manipulation and retrieval. Networking concepts such as client-server architecture will be applied to allow multiple users to access the system simultaneously. Optionally, a simple user interface may be developed to interact with the database.
Expected Deliverables:	• Complete and functional database system • Entity-Relationship Diagram (ERD) • Database schema and table design • SQL scripts (DDL & DML) • Project documentation/report • Final presentation • User interface for system interaction